



Fineva: A Benchmark for Evaluating Finance-Focused LLM Agents



Researcher: XXXXXX Supervisor: Prof. HO Tsung Yi (CSE)

1) Introduction

- Large Language Models like GPT-4 have revolutionized various fields, including finance, by excelling in tasks such as text classification, basic calculation, and code generation.
- Their ability to process and generate human-like text, coupled with rapid response times, makes them invaluable in finance, where accuracy and timeliness are crucial.
- Have significant potential for classifying news headlines, recognizing named entities, predicting stock movements, and providing basic financial analysis.

2) Existing Benchmarks

Existing financial benchmarks: FinBen, PIXIU, FinanceBench, BizBench.

Limitations:

- They focus on narrow Financial NLP tasks (e.g., information extraction, question answering).
- They lack real-world task evaluation and diverse skill assessment, resulting in no standardized framework for evaluating LLMs in real financial practice.
- They do not include real-time data or datasets that are actually used by businesses.

Benchmark	Language	Task	RFD	RFS	TA	QA	AMC	RM	PP
CFBenchmark	Chinese	7	x	x	3	x	x	x	x
Fin-Eva	Chinese	1	x	x	x	1	x	x	x
PIXIU	English	8	x	x	3	1	x	1	x
FinanceBench	English	1	x	x	x	1	x	x	x
BizBench	English	5	x	x	x	2	x	x	x
FinBen	English	24	x	x	8	3	x	4	x
Fineva	English	24	6	8	11	3	4	3	1

3) Research Objectives

- Develop an evaluation framework to assess LLM’s performance in real financial tasks.
- Bridges the gap left by existing benchmarks by incorporating a diverse range of financial scenarios and real financial world data.

4) Proposed Methodologies

Our benchmark primarily assesses the essential skills required for various financial roles, with an emphasis on the positions in investment banking.

- First, we categorized the different divisions within the bank, summarizing the scope of their daily work and the core skills needed, as outlined in the table below.
- Subsequently, tasks were designed based on each skill point. Each model was evaluated on 24 distinct financial tasks, which included calculating financial metrics, product pricing, knowledge Q&A, named entity recognition, risk management, and more.
- These tasks were assessed using zero-shot prompting, aiming to determine whether the LLM could complete the tasks based on a few sentences of instruction.

5) Models

Models: Selected three of the most common and accessible large language models (LLMs) available today to evaluate their performance across various financial tasks using our custom benchmark.

These models are widely recognized for their significance in the field of artificial intelligence and natural language processing: **GPT-3.5**, **GPT-4**, and **Claude**.

Reference:
Xie, Q., Han, W., Chen, Z., Xiang, R., Zhang, X., He, Y., Xiao, M., Li, D., Dai, Y., Feng, D., Xu, Y., Kang, H., Kuang, Z., Yuan, C., Yang, K., Luo, Z., Zhang, T., Liu, Z., Xiong, G., ... Huang, J. (2024). *FinBen: A holistic financial benchmark for large language models*. arXiv. <https://doi.org/10.48550/arXiv.2402.12659>
Xie, Q., Han, W., Zhang, X., Lai, Y., Peng, M., Lopez-Lira, A., & Huang, J. (2023). PIXIU: A large language model, instruction data, and evaluation benchmark for finance. arXiv. <https://doi.org/10.48550/arXiv.2306.05443>
Islam, P., Kannappan, A., Kiela, D., Qian, R., Scherrer, N., & Vidgen, B. (2023). FinanceBench: A new benchmark for financial question answering. arXiv. <https://doi.org/10.48550/arXiv.2311.11944>

6) Results

Data	Task	GPT-3.5	GPT-4o	Claude
CompCorp(*)	Comparable Company Analysis	1	1	1
AcctCal(*)	Calculation of Accounting Metrics	1	1	1
IPO(*)	Understanding IPO Procedures	0.717	0.5	0.761
Prtn(*)	Ability to Calculate Portfolio Return Metrics	0.102	0.82	0.76
Prisk(*)	Ability to Calculate Portfolio Risk Metrics	0.125	0.92	0.83
AAPL Option Chains	Product Pricing	0.121	1	1
TradBen(*)	Ability to calculate trading benchmark	0.343	0.862	0.853
ExcRule(*)	Understanding on stock exchange rule	0.685	0.29	0.692
FIGSA (Ruan et al., 2022)	Text Sentiment analysis	0.685	0.635	0.685
FOMC (Shah et al., 2023)	Text Sentiment analysis	0.661	0.708	0.72
NewsMTSC (Hamborg & Donnay, EACL 2021)	Headline Sentiment analysis	0.692	0.788	0.769
ECTSum (Mukherjee et al., 2022)	Text analysis (Earnings Call Transcripts, financial reports)	0.542	0.896	0.77
FNS 2020 (Mukwazvure & Supreethi, 2015)	Text analysis (Earnings Call Transcripts, financial reports)	0.694	0.823	0.858
Flare-cfa	Finance Basic Knowledge QnA	0.62	0.86	0.90
XSum: (Narayan et al., EMNLP 2018)	Text analysis (News and extraction)	0.153	0.134	0.156
Parcel(*)	Value at risk calculation	0.89	1	1
NER (Alvarado et al., 2015)	Named entity	0.208	0.272	0.213
Compliance(*)	Understanding on investment related law	0.492	0.57	0.62
PnLcal(*)	Calculate profit and loss	1	1	1

- Model **GPT-4o** consistently outperformed **GPT-3.5** in complex financial tasks, especially in portfolio return and risk metrics.
- Model **Claude** demonstrated strong performance and often matching or slightly surpassing **GPT-4o**, particularly in tasks like calculating trading benchmarks and compliance understanding.
- All models performed equally well in straightforward tasks like Comparable Company Analysis and Accounting Metrics.
- Model **GPT-3.5** struggled with more advanced calculations, such as Portfolio Return Metrics and Product Pricing, achieving significantly lower scores.
- Sentiment analysis and text analysis tasks showed comparable performance among all models, with slight variations depending on the specific dataset.
- Model **Claude** and **GPT-4o** excelled in tasks requiring detailed financial knowledge and calculations, such as Value at Risk calculation.

7) Conclusion and Future Works

- Our evaluation framework for LLMs demonstrates that advanced models like GPT-4o excel in handling intricate financial analyses, while GPT-3.5 shows limitations.
- The findings highlight the transformative role of LLMs and Auto-GPT in the evolving financial landscape, necessitating ongoing research to fully harness their capabilities.
- These technologies enhance financial decision-making and investment strategies, emphasizing the need for further refinement to maximize their potential.
- The integration of Auto-GPT and prompt engineering in finance significantly improves stock price prediction, complex pricing models, and financial report generation.

